# Filtering the 2023-2024 dataset for Aphis fabae

## Table of contents


**Summary** 1

**Setup** 2
- Packages . . . 2
- Helper functions . . . 2

**Broad-sense heritability calculations** 2
- Raw dataset . . . 2
- Missing data . . . 3
- Remove observations with no adult alive . . . 3

**Final filtering: keep genotypes with at least 5 repetitions** 7


## Summary


Calculation of key metrics based on the 2023-2024 Aphis fabae phenotyping of the PEAMust collection.

1) Calculation of the broad-sense heritability ("repeatability") as a function of the number of biological replicates.
2) Calculation of the correlation between sessions.
3) Decision to keep N biological replicates: compromise between repeatability and number of genotypes/observations.

# Setup

## Packages

This analysis depends on the following packages:

```
suppressPackageStartupMessages(library("tidyverse"))
suppressPackageStartupMessages(library("lme4"))
suppressPackageStartupMessages(library("Hmisc"))
suppressPackageStartupMessages(library("corrplot"))
suppressPackageStartupMessages(library("emmeans"))
suppressPackageStartupMessages(library("patchwork"))
```

## Helper functions

Two helper functions are created:
- `calculate_broad_heritability()` giving the broad sense heritability ($H^2$) and the number of genotypes of a given dataset.
- `create_correlation_plot()` to calculate the correlations between experimental sessions.

```
source("../../broad_heritability_function.R")
source("../../create_correlation_plot.R")
```

```
Loading required package: glue
```

# Broad-sense heritability calculations

## Raw dataset

```
data250_afabae_00 <- read.csv("../../../../../../01.data/2023_screen_specifics_251genotypes/0
  mutate(ivip = as.factor(ivip)) %>%
  mutate(session = as.factor(session))

res <- calculate_broad_heritability(data = data250_afabae_00)
```

> **Tip**
>
> Conclusions:
> - Heritability is equal to 0.104.
> - There are 219 genotypes in total.
> - There are 2664 measurements in total.

### Missing data

Remove NA values.

```
data250_afabae_01 <- data250_afabae_00 %>% filter(!is.na(larvae_total_number))
data250_afabae_01$ivip <- droplevels(data250_afabae_01$ivip)

res2 <- calculate_broad_heritability(data = data250_afabae_01)
```

> **Tip**
>
> Conclusions:
> - Heritability is equal to 0.104.
> - There are 217 genotypes in total.
> - There are 2370 measurements in total.

```
create_correlation_plot(data250_afabae_01,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
```

### Remove observations with no adult alive

We chose to remove all data from plants on which no adult aphid was found. This decision was made to get rid of larvae installation bias and only assess the reproduction ability of the aphid.

When the adult aphid dies, the number of larvae will drop but only because the adult stayed spent less time on the plant and not because the plant was not suitable for the reproduction.

```
data250_afabae_02<- data250_afabae_01 %>% filter(adult_alive != "0") #remove if no adult
data250_afabae_02$ivip <- droplevels(data250_afabae_02$ivip)

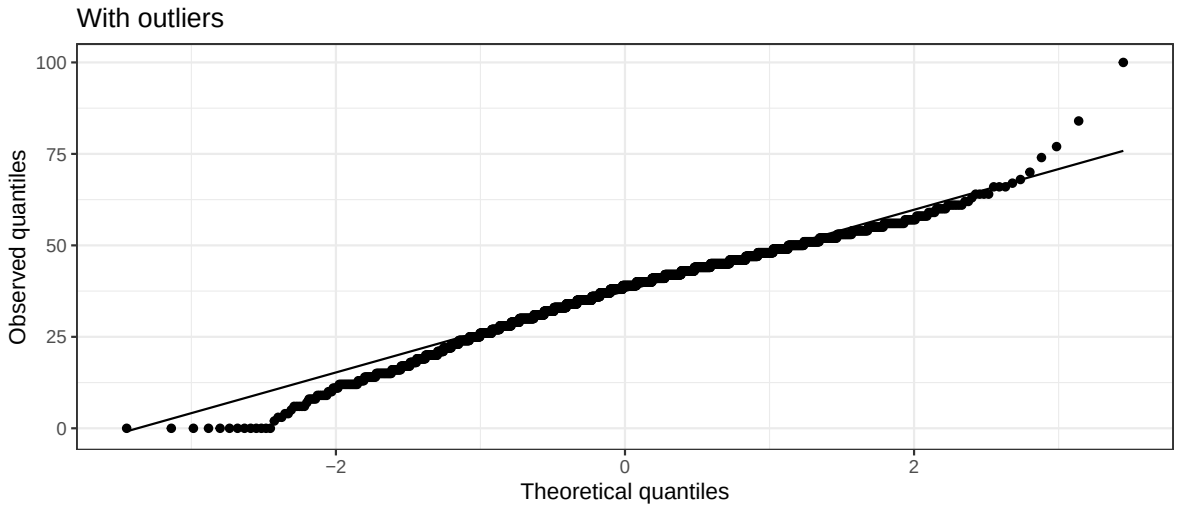
res3 <- calculate_broad_heritability(data = data250_afabae_02)
```

> **Note**
>
> Conclusions:
> - Heritability is equal to 0.296.
> - There are 217 genotypes in total.
> - There are 1768 measurements in total.

Heritability strongly increases. 602 measurements are lost. The number of genotype stays the same.

```
create_correlation_plot(data250_afabae_02,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
```

```
# With outliers
p_with <- data250_afabae_02 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("With outliers") +
  theme_bw()
p_with
```




We observ few outlier data on both side of the distribution. To remove these data we use the Interquartile range method (IQR). This will help inproving normality of the data.

```r
x<-1.5
IQR_value <- IQR(data250_afabae_02$larvae_total_number)
lower_bound <- quantile(data250_afabae_02$larvae_total_number, 0.25) - x * IQR_value
upper_bound <- quantile(data250_afabae_02$larvae_total_number, 0.75) + x * IQR_value

# Filter out outliers
data250_afabae_03 <- data250_afabae_02[data250_afabae_02$larvae_total_number >= lower_bound &

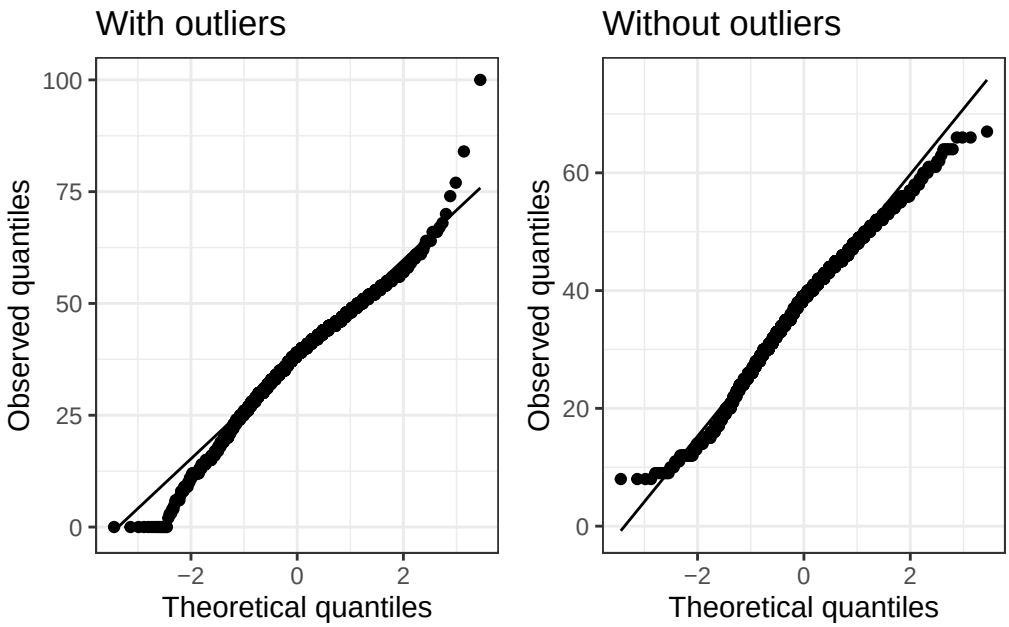
# Identify the rows that were removed (outliers)
removed_outliers <- data250_afabae_02[data250_afabae_02$larvae_total_number < lower_bound | d

res4 <- calculate_broad_heritability(data = data250_afabae_03)


#| fig-width: 8

# Without outliers
p_without <- data250_afabae_03 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Without outliers") +
  theme_bw()

p_with + p_without
```


With outliers
Observed quantiles
Theoretical quantiles
100
75
50
25
0
−2
0
2
Without outliers
Observed quantiles
Theoretical quantiles
60
40
20
0
−2
0
2

## Final filtering: keep genotypes with at least 5 repetitions

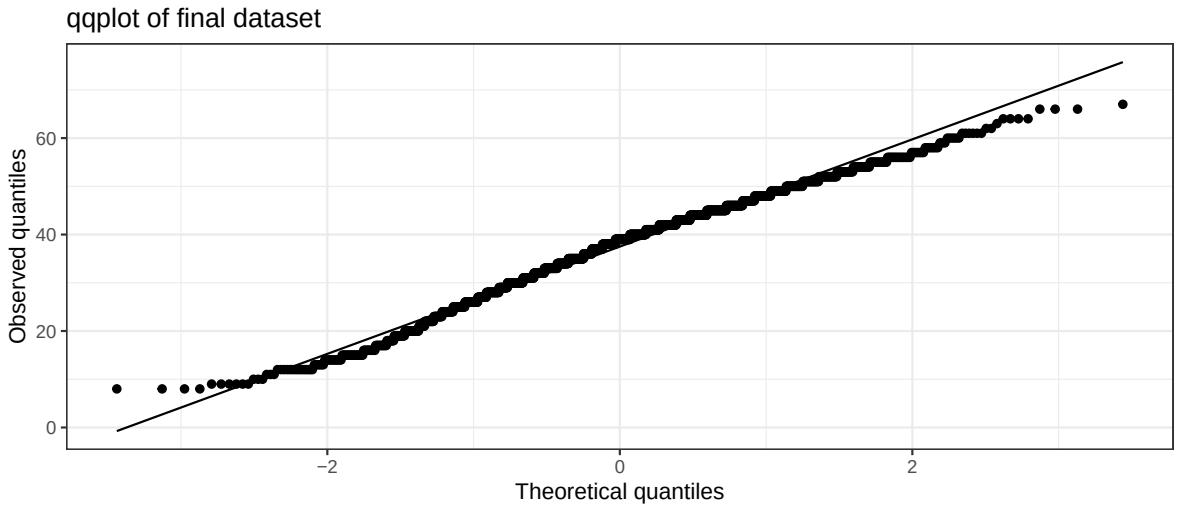
```
data250_afabae_04 <- data250_afabae_03 %>%
  group_by(ivip) %>%           # Group by genotype
  filter(n() >= 4) %>%             # Keep only groups with 6 or more repetitions
  ungroup()
data250_afabae_04$ivip <- droplevels(data250_afabae_04$ivip)

res5 <- calculate_broad_heritability(data = data250_afabae_04)
```

> **!** Important
>
> Conclusions:
> - Heritability is equal to 0.345.
> - There are 217 genotypes in total.
> - There are 1737 measurements in total.

```
create_correlation_plot(data250_afabae_04,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
qqnorm(data250_afabae_04$larvae_total_number)
```

```
p <- data250_afabae_04 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("qqplot of final dataset") +
  theme_bw()
p
```




### Write final results to .csv

```
write.csv(x = data250_afabae_04,
          file = "../../../01.repetability_choice_Nrep/01_A.fabae/final filters/aphis_fabae_
          quote = F, row.names = F)
```